

Write the sources of image noise. Describe the characteristics of noise normally present in digital image. Briefly describe the effects of different noise on image.

- *sources of image noise
- *characteristics of noise
- *effects of different noise

Sources of image noise.

The principal sources of noise in digital images arise during image acquisition and/or transmission.

Image Acquisition: Environmental factors such as light levels and sensor temperature introduce noise during the capture process.

Transmission: Atmospheric disturbances in wireless networks introduce noise during transmission.

Characteristics of Noise in Digital Images.

1. White Noise

- The Fourier spectrum of noise is constant across all frequencies.
- Noise is assumed to be independent of spatial coordinates.
- Noise is uncorrelated with respect to the image itself.
- Exception: spatially periodic noise.

2. Gaussian Noise

- Caused by electronic circuit noise, sensor noise due to poor illumination and/or high temperature.

3. Rayleigh Noise

- Commonly used in range imaging.
- PDF is skewed to the right.

4. Erlang Noise

- Used in laser imaging applications.

4. Exponential Noise

- Also associated with laser imaging.

5. Uniform Noise

- Least descriptive of real noise.
- Serves as the basis for numerous random number generators.

6. Impulse Noise

- Results from quick transients such as faulty switching.
- Appears as random black (pepper) and white (salt) pixels scattered in the image.

7. Periodic Noise

- Arises typically from electrical or electromechanical interference during image acquisition.
- It is a type of spatially dependent noise.
- Can be reduced significantly via frequency domain filtering.

Effects of Different Noise on Images

Gaussian Noise: Typically results from electronic circuit noise or sensor issues due to high temperatures or poor illumination.

Salt-and-Pepper Noise: Random black and white pixels caused by sharp transients or faulty switching during acquisition.

Periodic Noise: Electrical or electromechanical interference appearing as concentrated spikes in the image's Fourier transform.

Erlang and Exponential Noise: Commonly associated with laser imaging systems.

Erlang and Exponential Noise: Commonly associated with laser imaging systems.

Rayleigh: Skewed intensity distortion. Common in radar/range images

Uniform: Uniformly distributed random variation, gives a flat, grainy appearance.